- There are pockets of warm air or warm sector wedged between the forward and the rear cold air or cold sector. The warm air glides over the cold air and a sequence of clouds appear over the sky ahead of the warm front and cause precipitation.
- The cold front approaches the warm air from behind and pushes the warm air up. As a result, cumulus clouds develop along the cold front. The cold front moves faster than the warm front ultimately overtaking the warm front. The warm air is completely lifted up and the front is occluded (occluded front) and the cyclone dissipates.
- Normally, individual frontal cyclones exist for about three to ten days moving in a generally west to east direction.
- Precise movement of this weather system is controlled by the orientation of the polar jet stream in the upper troposphere.

95. **Statement 1: INCORRECT**—Tropical seas can be formed on seas only, whereas temperate cyclones can be formed on both land as well as oceans.

Statement 2: INCORRECT—Tropical cyclones occur in late summers and not all temperate cyclones occur in winter few will be formed in winters. Temperate cyclones are irregular.

Statement 3: INCORRECT— Tropical cyclones are limited to small area (100-500 kms in diameter) whereas Temperate cyclones cover large area (3002000 kms in diameter).

96. **Statement 1: CORRECT**—The Indian Ocean Dipole (IOD) is defined by the difference in sea surface temperature between two areas (or poles, hence a dipole) - a western pole in the Arabian Sea (western Indian Ocean) and an eastern pole in the eastern Indian Ocean south of Indonesia.

Statement 2: CORRECT—IOD develops in the equatorial region of Indian Ocean from April to May peaking in October.

Statement 3: INCORRECT—With a positive IOD winds over the Indian Ocean blow from east to west (from Bay of Bengal towards Arabian Sea). This results in the Arabian Sea (western Indian Ocean near African Coast) being much warmer and eastern Indian Ocean around Indonesia becoming colder and dry.

It was demonstrated that a positive IOD index often negated the